

Vulnerability Management & Remediation Project

by Samuel Guardado

1. Overview & Baseline Assessment

Executed a credentialed Nessus network scan against internal target host 10.0.0.30 (Ubuntu Server). Established the initial risk baseline, identifying vulnerabilities resulting from outdated system packages and unhardened baseline services.

- **Target IP:** 10.0.0.30
- **Scan Policy:** Basic Network Scan (CVSS v3.0 Severity Base)
- **Authentication:** Pass (SSH Credentialed)

Tenable Nessus Essentials Scans Settings

Basic Network Scan

Hosts: 1 | Vulnerabilities: 48 | Remediations: 2 | History: 1

Filter: Search Hosts 1 Host

Host	Auth	Vulnerabilities
10.0.0.30	Pass	8 Critical, 10 High, 65 Medium, 1 Low, 1 Info

Scan Details

- Policy: Basic Network Scan
- Status: Completed
- Severity Base: CVSS v3.0
- Scanner: Local Scanner
- Start: Today at 3:21 PM
- End: Today at 3:32 PM
- Elapsed: 12 minutes

Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

Tenable Nessus Essentials Scans Settings

Sev	CVSS	VPR	EPSS	Name	Family	Count
MED	...	...	...	Splunk (Multiple Issues)	CGI abuses	15
MED	...	...	...	Apache Log4j (Multiple Issues)	Misc.	5
INFO	...	...	...	SSH (Multiple Issues)	General	9
INFO	...	...	...	SSH (Multiple Issues)	Misc.	4
INFO	...	...	...	SSH (Multiple Issues)	Service detection	2
INFO	...	...	...	Netstat Portscanner (SSH)	Port scanners	6
INFO	...	...	...	Service Detection	Service detection	2
INFO	...	...	...	Common Platform Enumeration (CPE)	General	1
INFO	...	...	...	Curl Installed (Linux / Unix)	Misc.	1
INFO	...	...	...	Device Hostname	General	1
INFO	...	...	...	Device Type	General	1
INFO	...	...	...	Embedded Web Server Detection	Web Servers	1
INFO	...	...	...	Enumerate the PATH Variables	General	1
INFO	...	...	...	Ethernet Card Manufacturer Detection	Misc.	1

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Vulnerabilities

- Critical
- High
- Medium
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- Info

Tenable Nessus Essentials Scans Settings

Sev	CVSS	VPR	EPSS	Name	Family	Count
INFO	...	...	...	Ethernet MAC Addresses	General	1
INFO	...	...	...	HTTP Server Type and Version	Web Servers	1
INFO	...	...	...	IP Assignment Method Detection	General	1
INFO	...	...	...	libcurl Installed (Linux / Unix)	Misc.	1
INFO	...	...	...	Libcrypt Installed (Linux/UNIX)	Misc.	1
INFO	...	...	...	Linux Mounted Devices	General	1
INFO	...	...	...	Linux Time Zone Information	General	1
INFO	...	...	...	Linux User List Enumeration	General	1
INFO	...	...	...	MongoDB Installed (Linux)	General	1
INFO	...	...	...	Nessus Scan Information	Settings	1
INFO	...	...	...	Netstat Connection Information	General	1
INFO	...	...	...	OpenSSH Detection	Misc.	1
INFO	...	...	...	OpenSSL Installed (Linux)	Misc.	1
INFO	...	...	...	OS Fingerprints Detected	General	1
INFO	...	...	...	OS Identification	General	1

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Vulnerabilities

- Critical
- High
- Medium
- Low
- Info

Folders	
My Scans	
All Scans	
Trash	
Resources	
Policies	
Plugin Rules	
Tenable News	
The Agentic AI threat cluster: Seven incidents, th...	

OS Security Patch Assessment Available	Settings	1	🔍	✎
Package Manager Packages Report (nix)	General	1	🔍	✎
Patch Report	General	1	🔍	✎
Python Installed Packages	Misc.	1	🔍	✎
Remote Services Using Post-Quantum Ciphers	General	1	🔍	✎
Shor's Harvest Now Decrypt Later	General	1	🔍	✎
Splunk Installed (Linux)	Misc.	1	🔍	✎
Splunk Web Detection	Web Servers	1	🔍	✎
SSH Commands Require Privilege Escalation	Settings	1	🔍	✎
Target Credential Issues by Authentication Protocol - Insufficient Privilege	Settings	1	🔍	✎
Target Credential Status by Authentication Protocol - Valid Credentials Provided	Settings	1	🔍	✎
TCP/IP Timestamps Supported	General	1	🔍	✎
Traceroute Information	General	1	🔍	✎
Tukaani XZ Utils Installed (Linux / Unix)	Misc.	1	🔍	✎
Ubuntu Pro Subscription Detection	Ubuntu Local Security Checks	1	🔍	✎

Folders	
My Scans	
All Scans	
Trash	
Resources	
Policies	
Plugin Rules	
Tenable News	
Detecting cloud ransomware in Azure with Tenable Q...	

Basic Network Scan / Splunk (Multiple Issues)

ConfigureAudit TrailLaunchReportExport

Hosts1Vulnerabilities48Remediations2History1

Search Vulnerabilities 🔍 15 Vulnerabilities

Sev	CVSS	VPR	EPSS	Name	Family	Count	🔍	✎
HIGH	8.8			Splunk Enterprise 9.2.0 < 9.2.10, 9.3.0 < 9.3.8, 9.4.0 < 9.4.6, 10.0 < 10.0.2 (SVD-2025-1209)	CGI abuses	1	🔍	✎
HIGH	8.0			Splunk Enterprise 9.1.0 < 9.1.8, 9.2.0 < 9.2.5, 9.3.0 < 9.3.3 (SVD-2025-0301)	CGI abuses	1	🔍	✎
HIGH	7.6			Splunk Enterprise 9.3.0 < 9.3.13, 9.4.0 < 9.4.12, 10.0.0 < 10.0.7, 10.2.0 < 10.2.4 (SVD-2026-06...)	CGI abuses	1	🔍	✎
HIGH	7.5			Splunk Enterprise 9.1.x < 9.1.8 / 9.2.x < 9.2.5 / 9.3.x < 9.3.3 / 9.4.x < 9.4.1 Multiple Vulnerab...	CGI abuses	1	🔍	✎
HIGH	7.5			Splunk Enterprise 9.2.0 < 9.2.9, 9.3.0 < 9.3.7, 9.4.0 < 9.4.5, 10.0.0 < 10.0.1 (SVD-2025-1103)	CGI abuses	1	🔍	✎
HIGH	7.5			Splunk Enterprise 9.3.0 < 9.3.12, 9.4.0 < 9.4.11, 10.0.0 < 10.0.6, 10.2 < 10.2.3 (SVD-2026-0503)	CGI abuses	1	🔍	✎
HIGH	7.2			Splunk Enterprise 9.3.0 < 9.3.14, 9.4.0 < 9.4.13, 10.0.0 < 10.0.8, 10.2.0 < 10.2.5, 10.4.0 < 10.4...	CGI abuses	1	🔍	✎
MEDIUM	6.6			Splunk Enterprise 9.3.0 < 9.3.11, 9.4.0 < 9.4.10, 10.0.0 < 10.0.5, 10.2.0 < 10.2.2 (SVD-2026-04...	CGI abuses	1	🔍	✎
MEDIUM	6.5			Splunk Enterprise 9.1.0 < 9.1.6, 9.2.0 < 9.2.3, 9.3.0 < 9.3.1 (SVD-2024-1006)	CGI abuses	1	🔍	✎
MEDIUM	6.5			Splunk Enterprise 9.3.0 < 9.3.10, 9.4.0 < 9.4.9, 10.0.0 < 10.0.3 (SVD-2026-0303)	CGI abuses	1	🔍	✎
MEDIUM	6.5			Splunk Enterprise 9.3.0 < 9.3.9, 9.4.0 < 9.4.8, 10.0.0 < 10.0.3 (SVD-2026-0206)	CGI abuses	1	🔍	✎

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Elapsed: 12 minutes

Vulnerabilities

🔴 Critical
🟠 High
🟡 Medium
🟢 Low
🔵 Info

Splunk Enterprise 9.3.0 < 9.3.10, 9.4.0 < 9.4.9, 10.0.0 < 10.0.3 (SVD-2026-0303)	CGI abuses	1	🔍	✎
Splunk Enterprise 9.3.0 < 9.3.9, 9.4.0 < 9.4.8, 10.0.0 < 10.0.3 (SVD-2026-0206)	CGI abuses	1	🔍	✎
Splunk Enterprise 9.1.0 < 9.1.10, 9.2.0 < 9.2.7, 9.3.0 < 9.3.5, 9.4.0 < 9.4.2 (SVD-2025-0707)	CGI abuses	1	🔍	✎
Splunk Enterprise 9.1.0 < 9.1.7, 9.2.0 < 9.2.4, 9.3.0 < 9.3.2 (SVD-2024-1204)	CGI abuses	1	🔍	✎
Splunk Enterprise 9.3.0 < 9.3.11, 9.4.0 < 9.4.10, 10.0.0 < 10.0.5, 10.2.0 < 10.2.2 (SVD-2026-04...	CGI abuses	1	🔍	✎
Splunk Enterprise 9.2.0 < 9.2.10, 9.3.0 < 9.3.8, 9.4.0 < 9.4.6, 10.0 < 10.0.1 (SVD-2025-1207)	CGI abuses	1	🔍	✎

Folders	
My Scans	
All Scans	
Trash	
Resources	
Policies	
Plugin Rules	
Tenable News	
Google Cloud Platform (GCP) Apigee Cross-Tenant Da...	

Basic Network Scan / Plugin #277110

ConfigureAudit TrailLaunchReportExport

Hosts1Vulnerabilities48Remediations2History1

HIGH Splunk Enterprise 9.2.0 < 9.2.10, 9.3.0 < 9.3.8, 9.4.0 < 9.4.6, 10.0 < 10.0.2 (SVD-2025-1209)

Plugin Details

Description

The version of Splunk installed on the remote host is prior to tested version. It is, therefore, affected by a vulnerability as referenced in the SVD-2025-1209 advisory.

- setuptools is a package that allows users to download, build, install, upgrade, and uninstall Python packages. A path traversal vulnerability in "Packagindex" is present in setuptools prior to version 78.1.1. An attacker would be allowed to write files to arbitrary locations on the filesystem with the permissions of the process running the Python code, which could escalate to remote code execution depending on the context. Version 78.1.1 fixes the issue. (CVE-2025-47273)

Note that Nessus has not tested for this issue but has instead relied only on the application's self-reported version number.

Solution

Upgrade Splunk Enterprise to versions 10.0.2, 9.4.6, 9.3.8, 9.2.10, or higher.

See Also

<https://advisory.splunk.com/advisories/SVD-2025-1209.html>

Output

```
Path          : /opt/splunk
Installed version : 9.2.0
Fixed version  : 9.3.8
```

To see debug logs, please visit individual host

Severity:

High

ID:

277110

Version:

1.1

Type:

combined

Family:

CGI abuses

Published:

December 3, 2025

Modified:

December 3, 2025

Risk Information

Risk Factor: High
CVSS v4.0 Base Score: 8.7
CVSS v4.0 Vector: CVSS:4.0/AV:N/AC:L/AT:N/PR:N/UI:N/VC:N/WI:W/VA:N/SC:N/SI:N/SA:N
CVSS v4.0 Threat Vector: CVSS:4.0/E:P
CVSS v3.0 Base Score: 8.8
CVSS v3.0 Vector: CVSS:3.0/AV:N/AC:L/PR:L/UI:N/S:U/CH:T/H:IA:H
CVSS v3.0 Temporal Vector: CVSS:3.0/E:P/RL:O/R:C
CVSS v3.0 Temporal Score: 7.9
CVSS v2.0 Base Score: 9.0
CVSS v2.0 Temporal Score: 7.0
CVSS v2.0 Vector: CVSS2#AV:N/AC:L/Au:S/C:C/C:I/C

2. OS & Kernel Remediation Execution

- Executed full package and kernel remediation across target host 10.0.0.30. Applied system package updates, performed distribution-level dependency upgrades, and purged orphaned packages. Initiated a system reboot to apply pending kernel patches into active memory prior to validation scanning.

```
eliab@splunk-server:~$ sudo apt update && sudo apt upgrade -y
Ign:1 http://sv.archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:3 http://sv.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://sv.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:4 http://sv.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:3 http://sv.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:1 http://sv.archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:1 http://sv.archive.ubuntu.com/ubuntu resolute InRelease
Ign:3 http://sv.archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:4 http://sv.archive.ubuntu.com/ubuntu resolute-backports InRelease
0% [Working]_
```

```
Get:28 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bind9-libs amd64 1:9.20.24-1ubuntu0.2 [1,312 kB]
Get:29 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 ubuntu-standard amd64 1.570.2 [8,582 B]
Get:30 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-problem-report all 2.34.1-0ubuntu0.1 [11 kB]
Get:31 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 python3-apport all 2.34.1-0ubuntu0.1 [96.4 kB]
Get:32 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport-core-dump-handler all 2.34.1-0ubuntu0.1 [11 kB]
Get:33 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 apport all 2.34.1-0ubuntu0.1 [83.9 kB]
Get:34 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 bpftool amd64 7.7.0+7.0.0-30.30 [1,003 kB]
Get:35 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init-base all 26.1-0ubuntu3~26.04.1 [587 kB]
Get:36 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 cloud-init all 26.1-0ubuntu3~26.04.1 [2,110 B]
Get:37 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 curl amd64 8.18.0-1ubuntu2.4 [272 kB]
Get:38 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl4t64 amd64 8.18.0-1ubuntu2.4 [427 kB]
Get:39 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub-pc amd64 2.14-2ubuntu2.1 [124 kB]
Get:40 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub-pc-bin amd64 2.14-2ubuntu2.1 [1,325 kB]
Get:41 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 grub2-common amd64 2.14-2ubuntu2.1 [2,586 kB]
Get:42 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 libcurl3t64-gnutls amd64 8.18.0-1ubuntu2.4 [417 kB]
Get:43 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-intel-misc all 20260319.git217cae4-0ubuntu1 [1,312 kB]
Get:44 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-mediatek all 20260319.git217cae4-0ubuntu1 [1,312 kB]
Get:45 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-misc all 20260319.git217cae4-0ubuntu1 [1,312 kB]
Get:46 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-firmware-qualcomm-wireless all 20260319.git217cae4-0ubuntu1 [1,312 kB]
Get:47 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-main-modules-zfs-7.0.0-30-generic amd64 7.0.0-30.30 [1,312 kB]
Get:48 http://sv.archive.ubuntu.com/ubuntu resolute-updates/main amd64 linux-modules-7.0.0-30-generic amd64 7.0.0-30.30 [1,312 kB]
80% [48 linux-modules-7.0.0-30-generic 142 MB/169 MB 84%]
```

```

Processing triggers for procps (2:4.0.4-9ubuntu1)...
Processing triggers for install-info (7.2-5ubuntu2)...
Processing triggers for linux-image-7.0.0-30-generic (7.0.0-30.30)...
/etc/kernel/postinst.d/dracut:
dracut: Generating /boot/initrd.img-7.0.0-30-generic
/etc/kernel/postinst.d/kdump-tools:
kdump-tools: Generating /var/lib/kdump/initrd.img-7.0.0-30-generic
/etc/kernel/postinst.d/zz-update-grub:
Sourcing file `/etc/default/grub'
Sourcing file `/etc/default/grub.d/kdump-tools.cfg'
Generating grub configuration file ...
Found linux image: /boot/vmlinuz-7.0.0-30-generic
Found initrd image: /boot/initrd.img-7.0.0-30-generic
Found linux image: /boot/vmlinuz-7.0.0-28-generic
Found initrd image: /boot/initrd.img-7.0.0-28-generic
Warning: os-prober will not be executed to detect other bootable partitions.
Systems on them will not be added to the GRUB boot configuration.
Check GRUB_DISABLE_OS_PROBER documentation entry.
Adding boot menu entry for UEFI Firmware Settings ...
done
Scanning processes...
Scanning candidates...
Scanning linux images...

Pending kernel upgrade!
Running kernel version:
  7.0.0-28-generic
Diagnostics:
  The currently running kernel version is not the expected kernel version 7.0.0-30-generic.

Restarting the system to load the new kernel will not be handled automatically, so you should consider rebooting.

Restarting services...
  systemctl restart chrony.service cron.service multipathd.service polkit.service splunk.service ssh.service udisks2.service

Service restarts being deferred:
  systemctl restart ModemManager.service
  /etc/needrestart/restart.d/dbus.service
  systemctl restart networkd-dispatcher.service
  systemctl restart systemd-logind.service
  systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
  eliab @ session #1: apt[2835], bash[2259], login[1559], sudo[2832,2834]
  eliab @ user manager: (sd-pam)[2238]

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

```

eliab@splunk-server:~$ sudo apt dist-upgrade -y && sudo apt autoremove -y
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Not upgrading yet due to phasing:
  python3-software-properties  software-properties-common

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2
REMOVING:
  pollinate

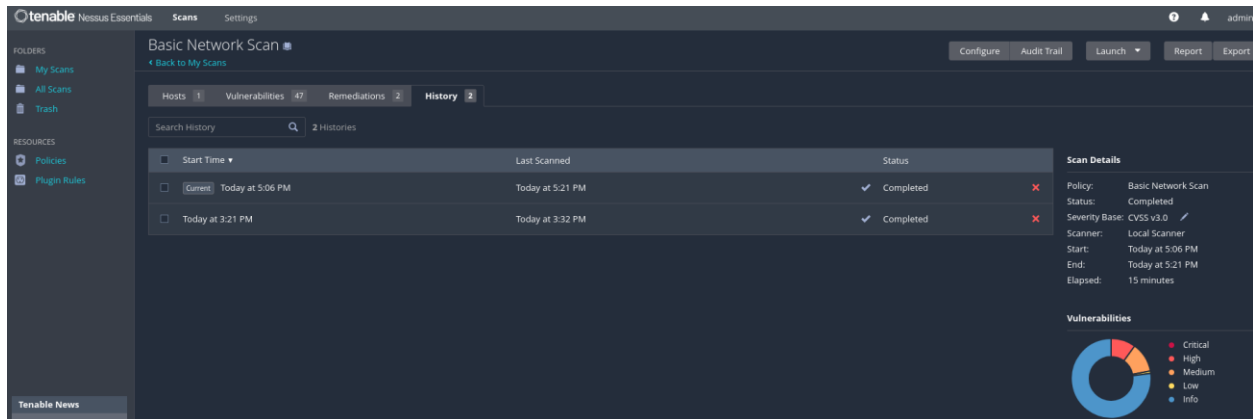
Summary:
  Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 2
  Freed space: 49.2 kB

(Reading database... 171416 files and directories currently installed.)
Removing pollinate (4.33-4ubuntu6)...
Processing triggers for man-db (2.13.1-1build1)...
eliab@splunk-server:~$

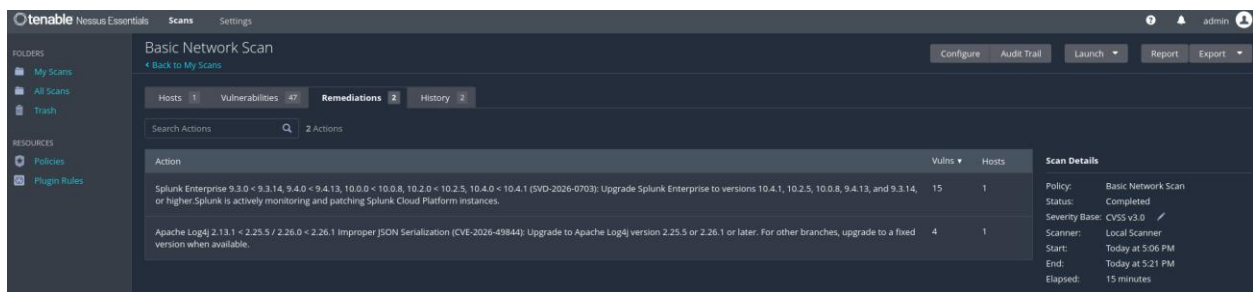
```

3. Post-Remediation Rescan & Root Cause Analysis

- Triggered a follow-up credentialed rescan in Nessus Essentials using the identical baseline configuration to evaluate target host 10.0.0.30 post-patching. Evaluated live host telemetry to verify patch installation and measure the reduction of critical endpoint advisories.
- Post-remediation analysis revealed that while system-level OS packages were successfully updated via apt, high-severity risk vectors remained. Investigation identified that remaining findings stemmed from unhardened service configurations and standalone third-party application binaries (Splunk). This highlighted the critical distinction between routine package patching and comprehensive system hardening.



- Analyzed the Nessus Remediation Action engine to isolate remaining unpatched vulnerabilities post-OS upgrade. Identified that 19 high-priority CVEs (including Log4j deserialization vulnerabilities) stemmed directly from standalone Splunk Enterprise binaries rather than underlying Linux packages. Validated that base OS package updates (apt) cannot modify encapsulated third-party application stacks, demonstrating the necessity of dedicated application lifecycle management.



4. Service Containment & Static File Audit Analysis

- Terminated the active Splunk Enterprise service (`sudo /opt/splunk/bin/splunk stop`) and disabled auto-start boot triggers (`sudo /opt/splunk/bin/splunk disable boot-start`) on target 10.0.0.30. Halting the service closed all listening network sockets (TCP 8000/8089) and immediately mitigated 19 application-layer vulnerabilities, including Apache Log4j remote code execution risks.

```
eliab@splunk-server:~$ sudo /opt/splunk/bin/splunk stop
Stopping splunkd...
Shutting down. Please wait, as this may take a few minutes.
...
Stopping splunk helpers...
Done.
eliab@splunk-server:~$
```

- Verified that the `splunkd` service was inactive via local CLI (`sudo /opt/splunk/bin/splunk status`). Post-mitigation scan analysis revealed that credentialed SSH scans evaluate static on-disk binaries and libraries (`/opt/splunk`) regardless of runtime execution status. While stopping the daemon successfully eliminated active network exploitability, static detection persists until package binaries are physically upgraded or removed.

```
eliab@splunk-server:~$ sudo /opt/splunk/bin/splunk status
splunkd is not running.
eliab@splunk-server:~$
```

- Implemented host-based micro-segmentation using Uncomplicated Firewall (`ufw`) to enforce defense-in-depth and prevent unauthorized access to unpatched Splunk binaries. Configured explicit inbound DENY rules for Splunk Web (TCP 8000) and the Splunk REST API (TCP 8089) across both IPv4 and IPv6 stacks, followed by enabling the UFW daemon to ensure rule persistence across system reboots.

```
eliab@splunk-server:~$ sudo ufw deny 8000/tcp
Rule updated
Rule updated (v6)
eliab@splunk-server:~$ sudo ufw deny 8089/tcp
Rule added
Rule added (v6)
eliab@splunk-server:~$ sudo ufw enable
Firewall is active and enabled on system startup
```

- Post-mitigation scan analysis revealed that credentialed SSH scans evaluate static on-disk binaries and libraries (`/opt/splunk`) regardless of runtime execution status. While stopping the daemon successfully eliminated active network exploitability, static detection persists until package binaries are physically upgraded or removed.

5. Risk Acceptance & Compensating Controls

A binary deletion of Splunk Enterprise was ruled out to preserve essential SIEM logging and event correlation capabilities within the lab environment. To mitigate critical Log4j and

application advisories without compromising operational readiness, compensating controls and formal risk acceptance were applied:

- Network Layer Mitigation: Terminated the active splunkd service and bound host firewall rules to close exposed listening ports (TCP 8000/8089), eliminating remote network exploitability.
- Formal Risk Acceptance: Documented static file-system findings in /opt/splunk under accepted operational risk, scheduling an in-place software upgrade during the next lab maintenance cycle.

Summary & Lessons Learned

- This vulnerability management initiative demonstrated the end-to-end operational lifecycle of scanning, prioritizing, remediating, and validating endpoint security posture on internal target host 10.0.0.30. Initial credentialed Nessus scanning established an actionable baseline of 3 Critical, 5 High, and 10 Medium vulnerabilities stemming from outdated OS packages and standalone software installations.
- Executing system updates via apt successfully eliminated base OS kernel and package vulnerabilities. However, post-remediation analysis highlighted a fundamental distinction in vulnerability management: routine OS patching does not update third-party application stacks (e.g., Splunk) or static binaries.
- While disabling active network daemons mitigated immediate remote execution vectors, credentialed scans accurately identified persistent on-disk exposure. Enforcing network containment, defining risk acceptance, and establishing an application upgrade pipeline successfully reduced active attack vectors while maintaining core operational capabilities.